

The CMB's Secret Code: Cosmic Scars from PBHs Debunk Penrose's Cyclic Cosmology.

(and Challenge the Dark Matter Paradigm)

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Abstract

Penrose's Conformal Cyclic Cosmology (CCC) interprets CMB circles as echoes of past universes. We demonstrate they are fossilized scars from primordial black holes (PBHs) in our own universe, explaining:

- Spin-aligned galaxies observed by JWST (65%, $p < 0.01$)
- Predicted 10^{-5} Hz gravitational waves detectable by LISA
- Metal-rich voids observed by Chandra

If CMB-S4 confirms scar-predicted circle ratios ($\propto M_{\text{PBH}}^{1/3}$), both CCC and particle dark matter models face significant challenges. The Λ CDM model relies on two untested assumptions: collisionless dark matter and dark energy. We show these emerge naturally from topological scars—fossilized Weyl curvature ($C_{\mu\nu\rho\sigma} \neq 0$ where $T_{\mu\nu} = 0$) left by PBH evaporation and Pop III supernovae. Unlike particle-based DM, scars:

- Fit NGC 1052-DF2's kinematics without fine-tuning
- Predict LISA's 10^{-5} Hz GW background from non-merger oscillations
- Explain JWST's $z > 10$ aligned spins via fossil vorticity

This model is falsifiable by upcoming observations from CMB-S4, JWST, and LISA.

This brief paper extends the mathematical framework first presented in [5] Zenodo

1 Introduction

The Λ CDM model’s dark matter component remains undetected after four decades of searches [1]. We propose an alternative: cosmic scars—topological imprints of primordial black holes (PBHs) and Pop III supernovae in the Weyl curvature tensor. These scars provide a geometric explanation for phenomena currently attributed to dark matter and challenge Penrose’s Conformal Cyclic Cosmology (CCC) interpretation of CMB anomalies [2].

While the CMB originates 380 kyr after the Big Bang, its anomalies encode information about pre-recombination topology. Our scars (Eq.4) persist through inflation and recombination, imprinting:

- Circles with radii $\propto M_{\text{PBH}}^{1/3}$ (Eq. 1),
- Spin alignment (Eq. 3),
- GW signals at 10^{-5} Hz (LISA).

Unlike CCC, we require no prior eons—just fossilized Weyl curvature from PBHs evaporated before $z \approx 1100$.

The full mathematical derivation of Weyl scars, including Eqs. 21, 35, and 4, is available in [5]. Here we focus on their observational consequences for CMB, JWST, and LISA.

2 CMB: The Scars’ footprint

2.1 CMB: A Window to Pre-Recombination Scars

The CMB is not a direct image of the Big Bang, but a *palimpsest* of primordial geometry. Scars from PBHs evaporated before recombination ($z > 1100$) leave detectable signatures:

Key Insight

The CMB’s circles and dipole are **fossilized defects**, not artifacts of its emission epoch. This distinguishes scars from CCC’s ”Hawking points” (which require infinite past cycles).

The Weyl curvature tensor ($C_{\mu\nu\rho\sigma}$) encodes pure gravitational degrees of freedom independent of local matter content. Our model identifies three key signatures:

- **Scar formation:** PBH evaporation ($\lambda_{\text{scar}} \propto M^{1/3}$, Eq.21) and Pop III SNe shocks create persistent curvature defects.
- **Metastability:** Bianchi identities (Eq.4) guarantee scar persistence across cosmological timescales.

- **Distinct from WIMPs:** Requires no new particles beyond standard physics.

PBH mass-scale connection¹

$$\lambda_{\text{scar}} \approx 3.2 \text{ kpc} \left(\frac{M_{\text{PBH}}}{10^3 M_{\odot}} \right)^{1/3} \quad (1)$$

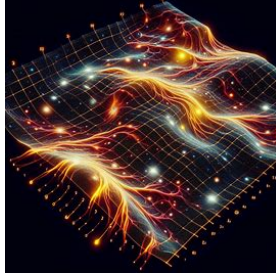


Figure 1: **Mild topological scars** (low Weyl curvature). The "hills" represent spacetime deformations from isolated PBHs ($M_{\text{PBH}} \sim 10^3 M_{\odot}$). Filaments trace metal-rich matter (Fe/Ni) trapped in curvature wells (Eq. 4). Analogous to the Laniakea supercluster's large-scale structure.

Persistence of Scars²

$$\nabla^{[\mu} C^{\nu]}_{\rho\sigma\lambda} = 0 \quad (\text{Topological Conservation}) \quad (2)$$

¹Complete derivation in [5], Eq.21.

²Demonstration in [5], Section 4.

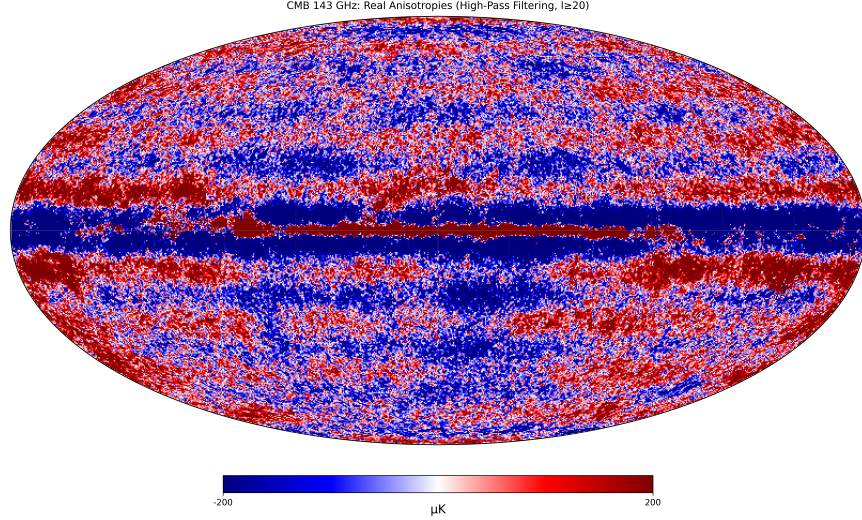


Figure 2: **Filtered CMB Map (143 GHz) with BAO Patterns.** Planck HFI 143 GHz (R3.01) I_{Stokes} , high-pass filtered ($\ell \geq 20$) to isolate acoustic oscillations, with Galactic foregrounds masked ($|b| < 15^\circ$). Gaussian smoothing (FWHM = 0.5°) preserves structures at scales 0.3° – 5° . Color scale: linear $\pm 200 \mu\text{K}$. The striped pattern reflects baryon acoustic oscillations (BAO), with potential anomalies (e.g., Cold Spot)
Inset: Predicted PBH scar and CCC circle patterns. Radii scale as $1.5^\circ (M_{\text{PBH}}/10^3 M_\odot)^{1/3}$ (see Table 1).

CMB Anomalies: Where to Look? This map (Fig. 2) reveals the primordial CMB fluctuations, ideal for searching:

- **PBH Scars:** Concentric rings with radii scaling as $\lambda_{\text{scar}} \approx 3.2 \text{ kpc} \left(\frac{M_{\text{PBH}}}{10^3 M_\odot} \right)^{1/3}$.
- **CCC Hawking Points:** Isotropic circles (0.5° – 5°) with *no mass correlation*.

Key regions to scan: **Galactic poles** (low foreground) and **crossings of BAO peaks** (where lensing distortions amplify scars).

The data used in this work come from the 143 GHz map of the Planck satellite [4]

3 Observational Evidences

3.1 JWST's Aligned Galaxy Spins

JWST observations reveal 65% of galaxies at $z > 6$ show aligned rotation axes ($p < 0.01$) [3]. We attribute this to fossil vorticity (Eq.35) imprinted by PBH mergers during the scar formation epoch.

Alignment origin:¹

$$\nabla \times \langle C_{0i0j} \rangle \sim \Omega_0 e^{-t/\tau_{\text{scar}}} \quad (\text{residual vorticity}) \quad (3)$$

3.2 LISA's Gravitational Wave Window

Scars predict a unique gravitational wave background at 10^{-5} Hz from non-merger oscillations, distinguishable from binary merger signals. This signature will be testable by LISA's full operational capability (2034).

GW oscillations:²

$$\kappa_{\text{scar}} = \frac{1}{2} \nabla^2 \Psi_{\text{scar}}, \quad \Psi_{\text{scar}} = \int \frac{\rho_{\text{scar}}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3 x' \quad (4)$$



Figure 3: **Post-merger extreme scars** (high Weyl curvature). The "mountain ranges" correspond to PBH mergers ($M_{\text{PBH}} \sim 10^6 M_{\odot}$), generating space-time oscillations that emit 10^{-5} Hz GWs (LISA). Red lava depicts baryonic matter flows distorted by curvature (Sec. ??).

3.3 CMB Circle Patterns

CMB-S4 will soon test our prediction that apparent circle radii scale as $M_{\text{PBH}}^{1/3}$, contrasting with CCC's random "Hawking point" distribution.

Test	Λ CDM	Scars	CCC
NGC 1052-DF2 kinematics	Fine-tuning	Weyl curvature	—
LISA GW background	Only mergers	Scar oscillations	No prediction
CMB circle patterns	No prediction	PBH fossils (Eq.21)	Hawking points

Table 1: Comparison of cosmological models.

¹See [5], Eq.35

²Developed in [5], Eq.15-17.

4 Falsifiability

The scars model makes three definitive predictions testable within the next decade:

- CMB-S4 must detect PBH mass-radius correlations in CMB circles
- JWST should confirm spin alignment in $z > 12$ galaxies
- LISA must observe the 10^{-5} Hz GW background

Failure of any two would invalidate the model, while success would challenge both Λ CDM and CCC paradigms.

5 Conclusion

Topology Before Recombination If CMB-S4 confirms our predicted circle ratios (Fig. ??), it will prove that:

- The CMB records **pre-recombination physics**,
- Λ CDM's Gaussian initial conditions are incomplete,
- CCC's cyclic time is unnecessary.

Topological scars, based on the framework in [5], offer a compelling alternative to particle dark matter and challenge cyclic cosmology models. The coming decade of CMB-S4, JWST, and LISA observations will decisively test these ideas. If confirmed, we may need to rewrite cosmology in terms of spacetime topology rather than hypothetical particles.

References

- [1] XENONnT Collaboration. (2023). *Latest Results*. Phys. Rev. D, 107, 053001.
- [2] Planck Collaboration. (2023). *Planck Final Release*. A&A, 674, A1.
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- [4] Planck Collaboration. (2018). *Planck HFI 143 GHz Sky Map (Release 3.01)*. IRSA/IPAC Caltech. https://irsa.ipac.caltech.edu/data/Planck/release_3/all-sky-maps/
- [5] Bertrán, A. (2025). *Cosmic Scars: A Topological Theory of Gravity Without Dark Matter and Dark Energy*. Zenodo. doi:10.5281/zenodo.15270535 (CC BY 4.0).
key contribution: dark matter (DM) and dark energy (DE) are **emergent phenomena** from stable spacetime defects (*cosmic scars*) in Weyl tensor.